

MMVP Evaluation — SRF Results & LLaVA Translation Results

Qwen2.5-VL-3B-Instruct (full dataset, 150 pairs / 300 images)

Method	Pair Acc	Image Acc	Pairs Correct
Baseline (no SRF)	39.33%	68.00%	59/150
SRF (best config)	44.00%	70.67%	66/150
Δ	+4.67pp	+2.67pp	+7 pairs

Run command:

```
cd /volumes2/mlm/lmms-eval
conda run -n mllm env PYTHONPATH="/volumes2/mlm/lmms-eval/srf:/volumes2/mlm/lmms-eval/srf/saliency" \
    python my_analysis/autoresearch_mmvp/mmvp_eval.py 2>&1 | tee
/tmp/mmvp_run.log
```

Per-block breakdown (15-pair groups)

Block	Pairs	Acc	Representative question type
0–14	15	67%	Orientation/direction (wings open/closed, dog facing)
90–104	15	67%	Color & appearance (stripe color)
120–134	15	60%	Spatial/viewpoint (camera angle)
45–59	15	47%	Text on object (words on lightbar)
135–149	15	40%	Fine-grained appearance (shirt vs T-shirt)
15–29	15	33%	State/condition (flag blowing)
30–44	15	33%	OCR / text in image (letter J visible?)
60–74	15	33%	Color detail (white kernels?)
105–119	15	33%	Spatial relation (resting on palm?)
75–89	15	27%	Motion/spatial ← hardest block

SRF gains most on orientation and color blocks; weakest on motion and OCR.

Comparison Against Published Methods

Model	Method	Pair Acc	Source
GPT-4V	proprietary	38.0%	MMVP paper (2401.06209)
Gemini Pro Vision	proprietary	26.7%	MMVP paper
Qwen2.5-VL-3B + SRF	training-free	44.00%	ours
Qwen2.5-VL-3B	baseline	39.33%	ours
LLaVA-1.5-7B + VAR	training-free	9.33%	VAR (2503.03321)
ShareGPT4V-7B	baseline	6.67%	MMVP paper
LLaVA-1.5-13B	baseline	5.33%	MMVP paper
LLaVA-1.5-7B	baseline	3.33%	MMVP paper
InstructBLIP-7B	baseline	0.0%	MMVP paper

Note: VAR and ours use different base models. Direct comparison of relative gains: VAR +6.0pp (from 3.33%), SRF +4.67pp (from 39.33% — a much stronger ceiling).

Best SRF Config for Qwen2.5-VL-3B MMVP

File: `my_analysis/autoresearch_mmvp/srf.py`

Saliency

```
SALIENCY = {
    "clip_coarse_grid": 7,          # 7x7 coarse CLIP grid
    "clip_top_k_pct": 0.30,        # top 30% tokens = salient mask
    "clip_use_soft": True,         # soft continuous weights (not binary
mask)
    "clip_fallback_thresh": 0.20,  # if max_sim < 0.20, don't apply mask
}
```

Noun extraction: `extract_clip_noun(question, mode="mmvp")` — MMVP-specific regex handles A/B option phrasing.

Bias / Boost

```

BIAS = {
    "layer_start":      8,
    "layer_end":        15,          # fusion zone: layers 8-15 of 36 total
    (~22-42%)
    "head_top_k_pct":   0.20,        # top 20% vision-aware heads → 3 heads
    selected
    "boost_alpha":      4.0,          # attention logit boost for salient
    tokens
    "background_eps":    0.2,          # mild suppression of non-salient
    tokens
    "sys_beta":          0.10,        # suppress system-prompt token
    attention
    "bias_mode":         "additive_logit",
    "srf_apply_phase":   "both",      # apply at prefill AND generation
                                         # ← critical: MMVP A/B decision is at
    prefill
    "text_beta":         0.0,          # text suppression disabled
}

```

Key findings from autoresearch

What was tried	Result
phase="generation" only	0.3933 (same as baseline — prefill is what matters)
phase="both"	0.4400 ← best
boost_alpha=8.0	0.3867 — too strong, distorts attention
boost_alpha=4.0	0.4400 — sweet spot
boost_alpha=2.0	0.4200 — slightly weak
layers 8-19 (wider)	0.3933 — regression, too many late layers interfere
layers 8-12 (narrow)	0.4067 — worse than 8-15
layers 8-15	0.4400 — optimal
background_eps=0.0	0.4200 — mild suppression (0.2) better than none
head_top_k=0.10	0.4067 — too few heads
head_top_k=0.30	0.3867 — too many heads
head_top_k=0.20	0.4400 — optimal
clip_fallback_thresh=0.10	0.4333 — same (65 pairs)
noun_extract refactor	+0.007 pair acc — fixed 23 bad MMVP extractions

Architecture Comparison: Qwen2.5-VL-3B vs LLaVA-1.5-7B

Property	Qwen2.5-VL-3B	LLaVA-1.5-7B
LM layers	36	32
Attention heads	16 (2 KV — GQA)	32 (32 KV — full MHA)
Hidden size	2048	4096
Vision encoder	Internal Qwen2-ViT (dynamic res)	External CLIP ViT-L/14 @ 336px
Vision tokens	Dynamic (~60–910, spatial merged)	Fixed: 576 (24×24 grid)
Spatial merge	<code>spatial_merge_size=2</code>	1 (flat MLP2x projection)
Image token marker	<code><\ image_pad\ ></code> (repeating block)	<code><image></code> (1 marker) → -200 fill
Chat template	Qwen2.5 format	USER: <code><image>\nQ\n</code> ASSISTANT: <code></code>
Vision-LM fusion	Qwen-native cross-attention zone	CLIP features → MLP → LM input embeds

SRF Changes for LLaVA-1.5-7B

What transfers unchanged

- `boost_alpha=4.0` — attention logit scale, model-agnostic
- `background_eps=0.2` — background suppression ratio
- `layer_start=8, layer_end=15` — good warmstart (same range in 32-layer LLaVA)
- `head_top_k_pct=0.20` — same fraction; selects ~6 heads (vs 3 for Qwen)
- `phase="both"` — MMVP A/B decision still at prefill
- `sys_beta=0.10` — system prompt suppression
- `bias_mode="additive_logit"` — model-agnostic
- `clip_coarse_grid=7, clip_top_k_pct=0.30` — CLIP saliency side unchanged
- `qwen_attn_patch.py` — patches `F.softmax` globally, works for any PyTorch attention

Code changes required

1. Model loading

```
# Qwen
from transformers import Qwen2_5_VLForConditionalGeneration
model = Qwen2_5_VLForConditionalGeneration.from_pretrained(
    "Qwen/Qwen2.5-VL-3B-Instruct", torch_dtype=torch.bfloat16,
    device_map="auto", attn_implementation="eager")

# LLaVA
from transformers import LlavaForConditionalGeneration, AutoProcessor
model = LlavaForConditionalGeneration.from_pretrained(
    "llava-hf/llava-1.5-7b-hf", torch_dtype=torch.float16,
    device_map="auto", attn_implementation="eager")
```

2. Processor and chat template

```
# Qwen: uses qwen_vl_utils.process_vision_info
from qwen_vl_utils import process_vision_info
text = processor.apply_chat_template(msgs, tokenize=False,
    add_generation_prompt=True)
vis, _ = process_vision_info(msgs)
inputs = processor(text=[text], images=vis, return_tensors="pt")

# LLaVA: standard processor, no process_vision_info
prompt = f"USER: <image>\n{question}\nASSISTANT:"
inputs = processor(text=prompt, images=image, return_tensors="pt")
```

3. Image token range detection

```
# Qwen: find first/last <|image_pad|> token
IMAGE_TOKEN = "<|image_pad|>"
img_token_id = processor.tokenizer.convert_tokens_to_ids(IMAGE_TOKEN)
# img_start = first position of img_token_id, img_end = last position

# LLaVA: find contiguous block of IMAGE_TOKEN_INDEX=-200 in input_ids
IMAGE_TOKEN_INDEX = -200 # LLaVA's placeholder before expansion
# After model.generate() expands, image tokens are a contiguous 576-token
block
# Alternative: find expansion range by comparing input and expanded input
lengths
```

4. Grid dimensions for CLIP saliency

```
# Qwen: dynamic, uses spatial_merge_size=2
# clip_salience.get_grid_dims(inputs, spatial_merge_size=2)
```

```
# → grid varies per image resolution

# LLaVA: always 24x24 = 576 tokens, no spatial merge
grid_h, grid_w = 24, 24 # fixed for LLaVA-1.5 at 336px input
# OR: pass spatial_merge_size=1 to get_grid_dims
```

5. Head calibration

```
# Same logic, but use LLaVA processor/template for calib samples
# LLaVA has 32 heads; top 20% = 6-7 heads selected
```

6. PYTHONPATH / imports (same fix as Qwen)

```
PYTHONPATH="/volumes2/mlm/lmms-eval/srf:/volumes2/mlm/lmms-eval/srf/saliency"
```

Suggested warmstart config for LLaVA

```
BIAS = {
    "layer_start":      8,      # same as Qwen (proportionally ~25% of 32
layers)
    "layer_end":        15,     # same (47% of 32 layers) – reasonable
warmstart
    "head_top_k_pct":   0.20,   # 20% of 32 heads = ~6 heads
    "boost_alpha":      4.0,    # same starting point
    "background_eps":   0.2,    # same
    "sys_beta":         0.10,   # same
    "bias_mode":        "additive_logit",
    "srf_apply_phase":  "both",
}
```

Expected tuning needed: LLaVA has a much lower MMVP baseline (~3.33%) than Qwen (39.33%), so the optimal alpha/layer range may differ. The autoresearch loop for LLaVA should start from this warmstart and sweep `layer_start/end` , `boost_alpha` , and `head_top_k_pct` .

File Locations

File	Purpose
my_analysis/autoresearch_mmvp/mmvp_eval.py	Eval harness (DO NOT MODIFY)

File	Purpose
my_analysis/autoresearch_mmvp/srf.py	Best Qwen MMVP config
my_analysis/autoresearch_mmvp/results.tsv	Autoresearch history
my_analysis/autoresearch_mmvp/last_run.json	Per-sample results of last run
srf/noun_extract.py	Noun extraction (mmvp mode)
srf/saliency/clip_salience.py	CLIP saliency with get_grid_dims